

MixiMotion: One-Step Text-to-Motion Generation via Multi-Candidate Set Distillation

Hung Dinh¹[0009–0003–6278–1092], Binh Mai¹[0009–0009–9957–8702], Tran Quoc Bao Le¹[0009–0001–9024–5858], Lam Nguyen¹, and Cong Tran¹[0000–0001–9467–4978]

Posts and Telecommunications Institute of Technology 15000, Hanoi, Vietnam
{dinhhung15082004, binhmai2205, k100iltqbao}@gmail.com,
congtt@ptit.edu.vn

Abstract. Diffusion- and flow-matching-based text-to-motion models have recently demonstrated impressive capability in generating realistic and diverse 3D human motions from natural language descriptions. However, these approaches rely on multi-step iterative sampling during inference, resulting in high computational cost and significant latency, which limit their applicability in interactive and real-time scenarios. In contrast, one-step text-to-motion generation provides a promising alternative by synthesizing motions in a single forward pass, enabling substantially faster inference. Despite this advantage, existing one-step methods often suffer from a noticeable quality gap compared with iterative generative models, particularly in terms of motion realism and semantic consistency. To address this challenge, we propose MixiMotion, a novel one-step text-to-motion framework that improves generation quality while maintaining the efficiency of single-step inference. Our method integrates offline teacher-guided supervision with a progressive training curriculum that incrementally enhances motion realism, semantic alignment, and human motion preference. This training strategy allows the model to effectively learn high-quality motion priors without requiring costly iterative sampling during inference. Extensive experiments on the Structured Semantic Alignment Evaluation (SSAE) benchmark demonstrate that MixiMotion achieves state-of-the-art performance among existing one-step text-to-motion methods, attaining a semantic alignment score of 0.8802 while narrowing the performance gap with iterative approaches. Project page: .

Keywords: Text-to-motion · Knowledge distillation · One-step generation · Human motion synthesis · Semantic alignment

1 Introduction

Text-to-motion generation maps natural-language descriptions to temporally coherent 3D human motion and supports applications such as interactive authoring, animation, games, and embodied agents. HumanML3D [5] established a

large paired text–motion benchmark for training and evaluation. MDM [11] introduced a strong motion-space diffusion formulation, while MLD [2] performs diffusion in a compact motion latent space. Building on these advances, recent flow-matching systems have further improved semantic fidelity and motion quality. In particular, HY-Motion-1.0 [13] combines a large multimodal transformer with iterative flow matching to generate expressive motion from detailed prompts. This quality, however, comes with repeated network evaluations during sampling, which limits latency-sensitive and resource-constrained deployment.

Existing acceleration methods approach this problem from different directions. MotionHiFlow [6] reduces sampling cost through hierarchical flow matching, while MotionLCM [3] and MoFlow [4] demonstrate that text-conditioned motion can be generated with very few, or even one, network evaluation. Strict one-step generation remains challenging. The student must model a stochastic and multimodal map from text and noise without averaging distinct valid motions. At the same time, it must preserve fine-grained prompt semantics and physical properties such as foot contact, root stability, and coordinated locomotion. Pointwise imitation of a single teacher target provides limited coverage of this multimodality, whereas latent-space distance alone may not expose visible failures such as skating, weak steps, or unintended root drift. These challenges motivate the central question of this work: *Can a strict one-step student preserve the semantic alignment and perceptual motion quality of a large HY-Motion model using only offline cached teacher generations?*

We address this question with MixiMotion, a one-step text-to-motion generator trained from offline multi-step teacher caches. A frozen teacher is run once to store multiple motion candidates and text features per prompt, and the student maps Gaussian noise to one-step motions via bidirectional nearest-neighbor set matching, plus decoded body-space constraints. A progressive curriculum then refines gait and contact, semantic balance, and distance-based preferences without any online teacher evaluation.

We evaluate MixiMotion after removing all exact normalized-text matches between its teacher bank and the 1,969-prompt SSAE evaluation set. Under an all-frame native-video judge, MixiMotion obtains 0.8802 SSAE compared with 0.8856 for HY-Motion-1.0 full, retaining 99.4% of its score; the paired confidence interval does not resolve a significant difference between the two systems. MixiMotion uses 460.0M rather than 1.04B motion-generator parameters and one network evaluation per candidate rather than 50 solver evaluations. On a common RTX 5090 harness, this corresponds to an $89.2\times$ generation-only speedup and a $6.75\times$ end-to-end speedup over the full model. A blinded human study similarly rates overall motion quality at 4.33/5 for MixiMotion and 4.50/5 for HY-Motion-1.0 full, while MixiMotion scores above the evaluated compact baselines.

Our contributions are as follows:

- We formulate one-step HY-Motion distillation as bidirectional set matching between multiple student noise samples and multiple offline cached teacher

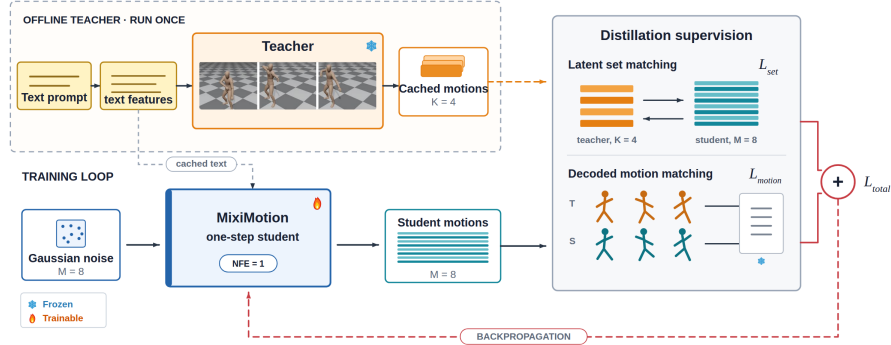


Fig. 1. Overview of MixiMotion distillation. A frozen HY-Motion-1.0-Lite teacher is run once to cache motion targets and text features. MixiMotion maps Gaussian noise to one-step motion candidates, supervised by latent set matching and decoded-motion matching; gradients update only the student.

motions, preserving multi-candidate supervision while requiring one network evaluation per generated candidate.

- We combine latent set matching with decoded motion constraints and study a progressive curriculum for motion quality, semantic balance, and preference refinement. Our development diagnostic reports the observed non-monotonic effects of these stages.
- We separate pre-filter development diagnostics from the exact-text-filtered primary evaluation, report paired bootstrap intervals, profile efficiency on common hardware, and conduct a blinded human study. HY-Motion-family results use matched rendering, whereas public-baseline comparisons are reported as contextual evidence under a common judge.

Figure 1 summarizes the offline teacher-cache construction and the one-step student supervision used to train MixiMotion.

2 Related Work

Text-to-motion generation. Diffusion and flow-based models have substantially improved the diversity and semantic fidelity of text-conditioned human motion. HumanML3D established a large text-motion benchmark [5]; T2M-GPT models discrete motion tokens autoregressively [14]; and MLD performs diffusion in a compact motion latent space [2]. MDM [11] introduced a strong motion-space diffusion formulation, while HY-Motion-1.0 scales flow-matching transformers and semantic evaluation [13]. Efficiency-oriented work includes latent consistency training in MotionLCM [3], one-step flow matching in MoFlow [4], and hierarchical flow matching in MotionHiFlow [6]. **MixiMotion instead learns a strict one-step student from offline multi-candidate teacher caches, using bidirectional set matching and a progressive curriculum without online teacher evaluation.**

Fast generative-model distillation. Progressive distillation reduces iterative samplers by repeatedly teaching a student to replace multiple teacher steps [9], while consistency models learn outputs that agree across points of a diffusion trajectory [10]. Our setting instead caches several complete teacher motions per prompt and learns a direct one-pass mapping. This avoids online teacher evaluation but requires the student to cover a multi-modal target set.

Set matching and preference learning. Implicit maximum likelihood estimation matches generated samples to nearby observations and offers an alternative to pointwise regression for multi-modal generation [7]. We use bidirectional nearest-neighbor coverage over student and teacher sets. In the final stage, we adapt the relative-preference principle of DPO [8] to distance-based motion scores with a frozen pre-preference student as reference. Unlike reward-model training, preference pairs are derived from deterministic physical statistics of cached teacher candidates.

3 Methodology

3.1 Teacher and MixiMotion

Let G_T denote a multi-step text-to-motion teacher and G_θ denote MixiMotion. Given a text prompt p , the teacher produces a set of candidate normalized motion latents

$$\mathcal{Y}_T(p) = \{y_1, \dots, y_K\}, \quad (1)$$

where each candidate is generated with a deterministic seed. MixiMotion receives the same text representation and predicts a motion latent in one network evaluation. Concretely, the student input consists of Gaussian noise tokens $z \in \mathbb{R}^{T \times D}$, token-level text context $C(p)$, a global text vector $v(p)$, and a temporal validity mask m . The student maps

$$\hat{y} = G_\theta(z, C(p), v(p), m), \quad z \sim \mathcal{N}(0, I), \quad (2)$$

where T is the motion length and D is the latent dimension. The implemented network encodes the noise tokens, text context, and text vector separately, forms a conditioning adapter from a masked noise summary and the text vector, passes motion and text features through double-stream and single-stream multimodal DiT blocks, and predicts the final normalized motion latent. A frozen teacher decoder is used only to recover joint trajectories for motion-quality analysis and video rendering.

3.2 Progressive Curriculum Distillation

The training recipe has four stages. Let $\mathcal{X}_\theta(p) = \{\hat{y}_1, \dots, \hat{y}_M\}$ be M student candidates generated from independent Gaussian noise tokens for the same prompt.

For two motion latents a, b with valid length ℓ , we use a masked Smooth-L1 distance

$$d_\delta(a, b; \ell) = \frac{1}{\ell D} \sum_{t=1}^{\ell} \sum_{d=1}^D \rho_\delta(a_{t,d} - b_{t,d}), \quad (3)$$

where ρ_δ is the Smooth-L1 penalty with transition point δ .

Base one-step distillation. The first stage learns the teacher distribution using a set-to-set IMLE-style objective. For any pairwise cost matrix $C \in \mathbb{R}^{K \times M}$, define the bidirectional nearest-neighbor reduction

$$\mathcal{B}(C) = \alpha_{\text{t2s}} \frac{1}{K} \sum_{j=1}^K \min_{i \leq M} C_{ji} + \alpha_{\text{s2t}} \frac{1}{M} \sum_{i=1}^M \min_{j \leq K} C_{ji}. \quad (4)$$

With $C_{ji}^{\text{lat}} = d_{\delta_{\text{lat}}}(y_j, \hat{y}_i; \ell)$, the latent set-matching loss is $\mathcal{L}_{\text{set}} = \mathcal{B}(C^{\text{lat}})$. The reported run uses $\alpha_{\text{t2s}} = 1$, $\alpha_{\text{s2t}} = 0.20$, and $\delta_{\text{lat}} = 0.35$.

Endpoint and sparse-trajectory costs use the same normalized representation. For a frame set S , let

$$d_\delta^S(a, b) = \frac{1}{|S|D} \sum_{t \in S} \sum_{d=1}^D \rho_\delta(a_{t,d} - b_{t,d}). \quad (5)$$

We set $S_{\text{end}} = \{\ell\}$ and $S_{\text{traj}} = \{1 + \text{round}(q(\ell - 1)/(P - 1))\}_{q=0}^{P-1}$ with $P = 4$, yielding C_{ji}^{end} and C_{ji}^{traj} by applying $d_{\delta_{\text{lat}}}^S$ to $(y_j, \hat{y}_i)$.

For decoded supervision, let $\Phi_r(y)$ be feature r extracted after the frozen body decoder, and let $\Omega_{r,j}(\ell)$ contain its valid temporal, joint, and coordinate entries. Each teacher-student pair has cost

$$C_{ji}^{(r)} = \frac{1}{|\Omega_{r,j}(\ell)|} \sum_{u \in \Omega_{r,j}(\ell)} \rho_{\delta_{\text{joint}}}([\Phi_r(y_j)]_u - [\Phi_r(\hat{y}_i)]_u). \quad (6)$$

Here, $r \in \mathcal{R}_{\text{base}}$ indexes body position, velocity, and acceleration; foot position and velocity; key-joint position and velocity; and foot-contact velocity. Velocities and accelerations are first and second temporal differences. Foot terms use joints (7, 8, 10, 11), and key-joint terms use (0, 10, 11, 15, 20, 21). The contact-velocity mask is computed from each teacher motion using speed threshold 0.02 and per-joint height thresholds (0.12, 0.12, 0.05, 0.05).

Crucially, the implementation sums decoded feature costs *before* the shared nearest-neighbor reduction. Thus, the complete base objective is

$$C_{ji}^{\text{body}} = \sum_{r \in \mathcal{R}_{\text{base}}} \lambda_r C_{ji}^{(r)}, \quad (7)$$

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{set}} + \lambda_{\text{end}} \mathcal{B}(C^{\text{end}}) + \lambda_{\text{traj}} \mathcal{B}(C^{\text{traj}}) + \mathcal{B}(C^{\text{body}}). \quad (8)$$

We use $\lambda_{\text{end}} = 0.10$, $\lambda_{\text{traj}} = 0.12$, and $\delta_{\text{joint}} = 0.04$. In the feature order above, the decoded weights are (0.20, 0.12, 0.04, 0.30, 0.25, 0.18, 0.12, 0.32). This stage gives MixiMotion a broad one-step motion prior while exposing errors that latent distance alone may hide.

Motion-quality curriculum. Stage 2 retains the latent, endpoint, and trajectory terms of $\mathcal{L}_{\text{base}}$, and the bidirectional reduction $\mathcal{B}$, but enlarges decoded matching. Let $\mathcal{P}$ be the training prompt pool and write

$$\begin{aligned}\mathcal{R}_{\text{gait}} &= \{\text{stride, lift, alt-contact, root-vel,} \\ &\quad \text{yaw-vel, arm-pos, arm-vel}\}, \\ \mathcal{R}_{\text{mq}} &= \mathcal{R}_{\text{base}} \cup \mathcal{R}_{\text{gait}}.\end{aligned}\tag{9}$$

Each $r \in \mathcal{R}_{\text{gait}}$ is a decoded feature map Φ_r on the frozen body trajectory $x = F_{\text{dec}}(y)$: Φ_{stride} is left-right foot separation in the xz -plane; Φ_{lift} is ground-relative foot height; $\Phi_{\text{alt-contact}}$ is a soft left/right contact pattern with height threshold $h_c = 0.05$ and temperature $\tau_c = 0.02$; $\Phi_{\text{root-vel}}$ and $\Phi_{\text{yaw-vel}}$ are horizontal root velocity and root-yaw velocity; arm channels use the arm-joint subset. Pair costs $C_{ji}^{(r)}$ for all $r \in \mathcal{R}_{\text{mq}}$ follow (6) with the same δ_{joint} . With nonnegative weights λ_r^{mq} , define

$$C_{ji}^{\text{mq}} = \sum_{r \in \mathcal{R}_{\text{mq}}} \lambda_r^{\text{mq}} C_{ji}^{(r)},\tag{10}$$

$$\mathcal{L}_{\text{mq}} = \mathcal{L}_{\text{set}} + \lambda_{\text{end}} \mathcal{B}(C^{\text{end}}) + \lambda_{\text{traj}} \mathcal{B}(C^{\text{traj}}) + \mathcal{B}(C^{\text{mq}}).\tag{11}$$

Here C^{end} and C^{traj} are as in the base stage, $\lambda_{\text{end}}, \lambda_{\text{traj}} > 0$ reweight those terms, and $\lambda^{\text{mq}} = (\lambda_r^{\text{mq}})_{r \in \mathcal{R}_{\text{mq}}}$ controls the decoded mix.

Lexical rules on prompt text define two lanes: $\mathcal{P}_{\text{loc}} \subseteq \mathcal{P}$ (locomotion; e.g., walk/run/step) and $\mathcal{P}_{\text{stat}} \subseteq \mathcal{P}$ (stationary/localized; e.g., reach/sit/wave, excluding locomotion tokens). Oversampling with integer factor $\kappa \geq 1$ builds a multiset by replicating each matched prompt κ times (sampling probability $\propto$ multiplicity). Residual prompts outside both lanes keep multiplicity one.

Stage 2 shares (11) across two subphases that differ only in the prompt measure and λ^{mq} :

- **Gait/body.** Sample $p \sim \text{Unif}(\mathcal{P})$ with the stage gait weights (reported run: $\alpha_{\text{t2s}} = 1, \alpha_{\text{s2t}} = 0.15$).
- **Hard locomotion.** Sample from $\mathcal{P}_{\text{loc}}$ expanded with $\kappa_{\text{loc}} = 6$, and raise the stride/lift/alt-contact weights to $(0.38, 0.85, 0.65)$ ($\alpha_{\text{s2t}} = 0.10$).

Both subphases supervise only against the offline cache $\mathcal{V}_T(p)$, targeting sliding, weak steps, and unstable lower-body motion.

Semantic-balanced curriculum. Stage 3 keeps set matching of the form (11) with stage-specific weights λ^{sem} , written $\mathcal{L}_{\text{mq}}^{\text{sem}}$, while rebalancing prompt mass: both $\mathcal{P}_{\text{loc}}$ and $\mathcal{P}_{\text{stat}}$ are oversampled with $\kappa_{\text{sem}} = 4$. For stationary prompts we add *student-only* root penalties (not teacher-student set matching). Let $\hat{\tau}_i \in$

$\mathbb{R}^{\ell \times 3}$ be the decoded root translation of student candidate $\hat{y}_i$ and define

$$\mathcal{L}_{\text{rd}}(p) = \frac{1}{M} \sum_{i=1}^M \frac{1}{\ell} \sum_{t=1}^{\ell} \rho_{\delta_{\text{joint}}}(\hat{\tau}_{i,t}^{[0,2]} - \hat{\tau}_{i,1}^{[0,2]}), \quad (12)$$

$$\mathcal{L}_{\text{rv}}(p) = \frac{1}{M} \sum_{i=1}^M \frac{1}{\ell-1} \sum_{t=1}^{\ell-1} \rho_{\delta_{\text{joint}}}(\hat{\tau}_{i,t+1}^{[0,2]} - \hat{\tau}_{i,t}^{[0,2]}). \quad (13)$$

Thus $\mathcal{L}_{\text{rd}}$ penalizes horizontal drift from the first frame and $\mathcal{L}_{\text{rv}}$ penalizes horizontal root velocity. The stage objective is

$$\mathcal{L}_{\text{sem}}(p) = \mathcal{L}_{\text{mq}}^{\text{sem}}(p) + \mathbf{1}[p \in \mathcal{P}_{\text{stat}}](\lambda_{\text{rd}} \mathcal{L}_{\text{rd}}(p) + \lambda_{\text{rv}} \mathcal{L}_{\text{rv}}(p)), \quad (14)$$

with reported $\lambda_{\text{rd}} = 0.80$ and $\lambda_{\text{rv}} = 0.30$. Anchored/other prompts are not a third oversampled lane; group-specific treatment enters only through the preference score $q(y, p)$ next.

Preference optimization. Direct Preference Optimization (DPO) [8] was introduced as a reference-relative preference objective that avoids fitting a separate reward model. We adapt that principle to motion generation: our score is a negative best-of- M motion distance rather than a language-model log probability. We first build preference pairs from teacher candidates. **For each prompt, the best and worst teacher candidates are selected according to the group-aware motion-quality score $q(y, p)$.** For locomotion prompts, q rewards foot lift, alternating contact, and stride while penalizing skating and contact velocity. For stationary and anchored prompts, q penalizes root displacement, root velocity, skating, and unnecessary foot lift. Other prompts use a mixed score that mildly rewards visible motion while penalizing instability. The winner and loser are

$$y^+ = \arg \max_{y_j \in \mathcal{Y}_T(p)} q(y_j, p), \quad y^- = \arg \min_{y_j \in \mathcal{Y}_T(p)} q(y_j, p). \quad (15)$$

MixiMotion is then optimized with this DPO-inspired objective plus supervised fine-tuning toward the preferred candidate. A frozen reference student G_{ref} is initialized from the model before preference training and evaluated on the same sampled noise tokens. Define the best-of- M score

$$s_{\theta}(y) = -\min_{i \leq M} d_{\delta}(\hat{y}_i, y; \ell), \quad s_{\text{ref}}(y) = -\min_{i \leq M} d_{\delta}(\hat{y}_i^{\text{ref}}, y; \ell). \quad (16)$$

The preference loss is

$$\mathcal{L}_{\text{DPO}} = -\log \sigma \left(\beta \left[(s_{\theta}(y^+) - s_{\theta}(y^-)) - (s_{\text{ref}}(y^+) - s_{\text{ref}}(y^-)) \right] \right), \quad (17)$$

and the reported preference stage uses

$$\mathcal{L}_{\text{SFT}} = \min_{i \leq M} d_{\delta}(\hat{y}_i, y^+; \ell), \quad (18)$$

$$\mathcal{L}_{\text{pref}} = \mathcal{L}_{\text{DPO}} + \lambda_{\text{sft}} \mathcal{L}_{\text{SFT}}. \quad (19)$$

This final stage sharpens MixiMotion without asking the preference loss to learn the full motion prior from scratch.

3.3 Training and Sampling Algorithms

Algorithm 1 Progressive curriculum distillation (PyTorch-like pseudocode).

Require: prompt bank $\mathcal{P}$; frozen HY-Motion-1.0-Lite teacher G_T and body decoder F_{dec} ; student G_θ ; EMA decay μ .

```

# Offline cache construction (run once)
1: Initialize the teacher bank  $\mathcal{D} \leftarrow \emptyset$ .
2: for  $p \in \mathcal{P}$  do
3:   Generate  $\mathcal{Y}_T(p) \leftarrow \{G_T(p; k)\}_{k=1}^K$ ,  $K = 4$ .
4:   Cache  $\mathcal{Y}_T(p)$  and the frozen text features.
5: end for
6: Remove rows that exactly match normalized SSAE text.
   # Stages 1-3: base, motion quality, and semantic balance
7: Initialize  $\theta_{\text{EMA}} \leftarrow \theta$ .
8: for phase  $s \in \{\text{base, gait/body, hard-locomotion, semantic}\}$  do
9:   for minibatch  $B \subset \mathcal{D}$  do
10:    Sample  $M = 8$  noises and compute  $\hat{y}_i \leftarrow G_\theta(z_i, C(p), v(p), m)$ .
11:    Decode when required and evaluate the phase loss  $\mathcal{L}_s$ .
12:    Update  $\theta$  with AdamW; update  $\theta_{\text{EMA}} \leftarrow \mu\theta_{\text{EMA}} + (1 - \mu)\theta$ .
13:   end for
14: end for
   # Stage 4: preference refinement
15: Build  $(p, y^+, y^-)$  using the group-aware score  $q(y, p)$ .
16: Set  $G_{\text{ref}} \leftarrow \text{stopgrad}(G_\theta)$ .
17: for preference minibatch do
18:   Sample  $M = 4$  shared noises; evaluate  $G_\theta$  and  $G_{\text{ref}}$  on the same noises.
19:   Compute best-of- $M$  scores and  $\mathcal{L}_{\text{pref}} = \mathcal{L}_{\text{DPO}} + 0.35\mathcal{L}_{\text{SFT}}$ .
20:   Update  $\theta$  and  $\theta_{\text{EMA}}$ .
21: end for
22: return  $G_{\theta_{\text{EMA}}}$ .

```

Algorithm 2 One-step MixiMotion sampling.

Require: EMA student $G_{\theta_{\text{EMA}}}$; prompt p ; requested length ℓ ; frozen text encoders and body decoder F_{dec} .

```

1: Encode  $p$  as  $C(p)$  and  $v(p)$ ; construct validity mask  $m$ .
2: Sample  $z \sim \mathcal{N}(0, I)$ .
3: Compute  $\hat{y} \leftarrow G_{\theta_{\text{EMA}}}(z, C(p), v(p), m)$ .
4: Decode  $x \leftarrow F_{\text{dec}}(\hat{y}, \ell)$ .
5: return motion  $x$ .

```

Algorithm 1 separates one-time teacher-cache construction from student optimization and makes the shared-noise reference comparison explicit. Algorithm 2

requires exactly one evaluation of $G_{\theta_{\text{EMA}}}$; text encoding and body decoding are outside the reported generator NFE.

Table 1. Selected implementation details for the reported MixiMotion training run.

Item	Value
Teacher for cached samples	HY-Motion-1.0-Lite
Teacher-bank rows after filtering	48,065
Exact normalized-text SSAE overlap	0
Teacher candidates per prompt K	4
Student candidates, base distillation M	8
Student candidates, preference stage M	4
Motion length	120 frames
Optimizer	AdamW
Base stage	18 epochs at 8.0×10^{-5}
Gait/body stage	10 epochs at 3.0×10^{-5}
Hard-locomotion stage	6 epochs at 2.0×10^{-5}
Semantic-balance stage	4 epochs at 1.0×10^{-5}
Preference stage	3 epochs at 2.0×10^{-6}
Batch size	4
DPO inverse temperature β	8.0
SFT weight λ_{sft}	0.35
Smooth-L1 transition δ	0.35
Network evaluations per candidate	1
EMA decay	0.999
Base-stage split	70% / 15% / 15% (train/val/test)
Preference-stage split	90% / 10% (train/val)
Precision	bfloat16

Figure 2 illustrates the curriculum as four consecutive training stages rather than as implementation-version checkpoints.

4 Experiments

4.1 Experimental Setup

Training data. The reported model uses a curated prompt bank collected from HY-Motion-1.0 examples, ViMoGen motion annotations [12], and manually filtered text-to-motion descriptions. For each prompt, HY-Motion-1.0-Lite generates four teacher candidates at a fixed length of 120 frames. Before the reported retraining run, we normalize text by stripping outer whitespace, converting to lowercase, and collapsing repeated whitespace. We remove all 1,935 cached rows whose normalized text matches an SSAE prompt; these rows cover 1,883 unique SSAE prompts. The resulting teacher bank contains 48,065 samples, and a post-filter audit finds zero exact normalized-text overlap with all 1,969 SSAE prompts. All curriculum stages and the preference bank are then rebuilt from this filtered

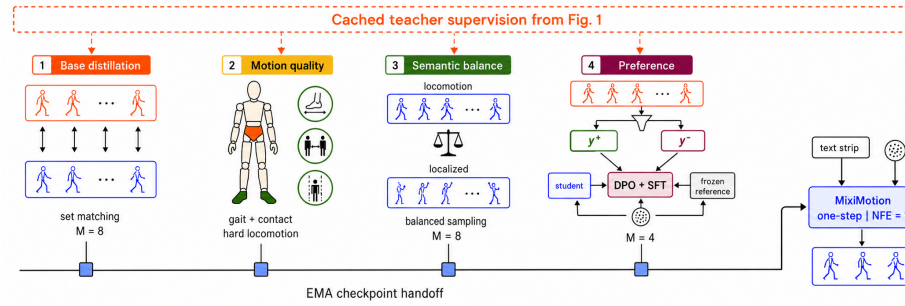


Fig. 2. Progressive curriculum used to train MixiMotion. Cached teacher supervision is reused across all stages; solid arrows denote EMA checkpoint handoff. The sequence does not imply monotonic improvement.

cache. This audit excludes exact text matches but does not rule out semantic paraphrases or near-duplicate descriptions.

Evaluation protocol. We evaluate on SSAE [13], which decomposes each text prompt into fine-grained yes/no questions about visible motion semantics. We render one video per SSAE index and per generator, producing 1,969 videos for each system and 7,876 in total. A VLM answers all questions for each video, and the score is the ratio of yes answers. Our native-video protocol passes all 120 frames of each rendered sequence to Qwen/Qwen3-VL-30B-A3B-Instruct-FP8 [1]; no temporal frame subsampling is used.

The primary comparison uses two motion generators:

- **MixiMotion:** our current one-step HY-Motion-1.0 model retrained from the exact-text-filtered teacher bank.
- **HY-Motion-1.0 full:** the public multi-step HY-Motion-1.0 model rendered with the same SSAE prompts and 120-frame length. Each benchmark run uses the deterministic seed recorded in its manifest.

For external context, we also run SSAE on two public text-to-motion baselines with available checkpoints: **MotionHiFlow** [6] and **MotionLCM** [3]. All systems use the same prompt-index mapping, 120-frame output length, SSAE questions, and native-video judge settings. MixiMotion and HY-Motion-1.0 full share the same 720×720 skeleton renderer, camera, and 30 FPS encoding (4 s). The public baselines use their native project renderers at 400×400 and 300×300 , respectively, and are encoded at 20 FPS (6 s). We do not temporally rescale these motions because doing so changes their native execution speed; consequently, public-baseline scores are contextual rather than strictly renderer- and duration-controlled comparisons.

For VLM judging, we use one all-frame native-video protocol:

- **Qwen/Qwen3-VL-30B-A3B-Instruct-FP8**, used for native-video evaluation with all 120 frames, identical SSAE questions, temperature zero, a

256-token response limit, and a strict yes/no JSON schema for every motion generator.

Strict structured decoding succeeded directly for 7,873 of the 7,876 videos. For three videos, the backend entered a whitespace loop after opening the JSON object. We reissued each question independently to the same model with the same 120-frame video, temperature, and yes/no instruction, changing only the output-format constraint. All four systems consequently contain 1,969 successful videos and the same 6,418 answered questions.

Evidence tiers and comparison scope. We distinguish development diagnostics from primary evidence. The cumulative curriculum checkpoints in Table 6 were inspected during method development and predate exact-text filtering. They reveal optimization trade-offs within one training chain, but they do not measure held-out generalization. The primary MixiMotion result in Table 2 is obtained only after rebuilding every training stage from the exact-text-filtered teacher bank. Exact filtering does not make SSAE an independently sourced benchmark, because semantic near-duplicates may remain.

We also distinguish matched and contextual model comparisons. MixiMotion and HY-Motion-1.0 full use the same latent family, renderer, camera, resolution, frame count, frame rate, prompts, and judge; this is the controlled primary comparison. MotionHiFlow and MotionLCM share the prompt indices, frame count, questions, and VLM settings, but retain their native renderers and frame rates. Their scores therefore provide external context rather than an estimate of a model-only effect. Paired confidence intervals quantify uncertainty for the videos produced by each protocol; they do not remove these renderer and timing confounds.

Efficiency profiling protocol. Parameter counts are computed from saved generation checkpoints and exclude text encoders, VLM judges, and video renderers. The two HY-Motion-family rows count the motion transformer. MotionHiFlow combines its flow generator and VAE, whereas MotionLCM combines its denoiser and VAE, so these external scopes are contextual rather than architecture-identical.

Every system is profiled on the same RTX 5090 with batch size 1, bfloat16 precision, and 120 output frames. We perform 20 unreported warm-up iterations followed by 100 timed iterations and synchronize CUDA around every measurement. Generation-only latency covers the native generator or solver. Model-native end-to-end latency additionally includes text encoding and each project decoder, while excluding video rendering and VLM judging. For MotionHiFlow, the public configuration sets `time_steps=12`, but its hierarchical scheduler executes 4, 4, and 6 transformer calls over the three scales; we therefore report 14 actual network evaluations. Its generation-only profile reuses the encoded prompt, whereas its end-to-end profile clears the text cache before every request and includes VAE decoding. HY-Motion-1.0 full and MixiMotion terminate at the same decoded HY-Motion representation. MotionHiFlow terminates

at decoded motion features, whereas MotionLCM additionally converts features to joints and copies them to CPU. We report mean, standard deviation, 95th percentile, and peak VRAM after resetting the CUDA peak-memory counter.

Human evaluation protocol. To complement the VLM-based assessment, we conduct a blinded human evaluation with 25 participants familiar with computer graphics or motion synthesis. The study covers 300 SSAE prompts per system. For each prompt, model outputs are shown in randomized pairs with model identities and left-right ordering hidden. Participants assign an independent five-point Likert score to each video for: (1) semantic alignment with the input prompt, (2) motion naturalness, and (3) overall motion quality, including foot-ground contact, stability, and visible artifacts. The reported values aggregate ratings over the 300 prompts for each system. We report the mean and standard error, and use Krippendorff’s alpha to summarize inter-rater agreement.

4.2 Experimental Results

Semantic evaluation. Table 2 reports the primary automatic comparison under the common all-frame judge defined in Section 4.1. MixiMotion and HY-Motion-1.0 full form the controlled same-family comparison; the public compact models provide context under the scope qualifications stated above.

Table 2. SSAE scores under the all-frame native-video Qwen/Qwen3-VL-30B-A3B-Instruct-FP8 judge. MixiMotion is retrained after removing exact normalized-text SSAE matches from its teacher bank. Every system has 1,969 successful videos and 6,418 answered questions. Higher is better; the best score in each row is in bold.

Category	MixiMotion (ours)	HY-Motion-1.0 full	Motion HiFlow	Motion LCM
Daily	0.9009	0.9075	0.8617	0.8065
Fitness	0.7487	0.7613	0.7717	0.7213
Game	0.9205	0.9215	0.9187	0.8977
Locomotion	0.9757	0.9757	0.9689	0.9621
Social	0.9332	0.9419	0.9452	0.9211
Sports	0.8430	0.8451	0.8274	0.7807
Overall	0.8802	0.8856	0.8764	0.8415

MixiMotion retains 99.4% of the HY-Motion-1.0 full score, with an absolute gap of 0.0055. It matches the full model on locomotion at the reported precision and is within 0.0021 on game and sports prompts. MotionHiFlow is strongest on fitness and social among the compact models, while MixiMotion has the numerically highest compact-model overall score. These small differences should be interpreted cautiously because the exact-text audit does not exclude semantic near-duplicates and the experiment uses a single deterministic VLM judge. A

paired video-level bootstrap with 10,000 resamples gives a 95% confidence interval of $[0.8687, 0.8915]$ for MixiMotion and $[0.8746, 0.8964]$ for HY-Motion-1.0 full. The paired difference (MixiMotion minus full) is -0.0055 with a 95% interval of $[-0.0121, 0.0011]$, which includes zero. MotionHiFlow’s paired difference is -0.0092 with interval $[-0.0186, 0.0002]$, whereas MotionLCM’s difference is -0.0441 with interval $[-0.0553, -0.0332]$. Thus, under this protocol, the VLM scores do not resolve a statistically clear difference between MixiMotion and the full baseline.

Efficiency and model size. Table 3 reports the controlled profile defined in Section 4.1. The two HY-Motion-family rows share the same decoded endpoint; cross-family end-to-end values retain the endpoint differences documented in the protocol.

Table 3. Controlled model-size and inference-cost profiling on 120-frame samples. Latency and memory use one RTX 5090/BF16 harness at batch size 1. External model-native endpoints differ as described in the text. Lower is better for every numeric column; best values are in bold.

<i>(a) Model complexity and memory</i>				
Model	Generator params (M)	Actual NFE	Peak VRAM (GiB)	
HY-Motion-1.0 full	1042.9	50	23.23	
MixiMotion (ours)	460.0	1	20.73	
MotionHiFlow	33.1	14	0.43	
MotionLCM	26.5	1	1.45	

<i>(b) Latency</i>				
Model	Generation mean $\pm$ std (ms)	Generation P95 (ms)	End-to-end mean $\pm$ std (ms)	End-to-end P95 (ms)
HY-Motion-1.0 full	829.58 $\pm$ 3.19	834.68	968.25 $\pm$ 3.69	974.94
MixiMotion (ours)	9.30 $\pm$ 0.39	9.94	143.55 $\pm$ 1.29	145.67
MotionHiFlow	77.32 $\pm$ 1.33	79.73	83.80 $\pm$ 1.17	85.82
MotionLCM	3.04 $\pm$ 0.24	3.35	13.94 $\pm$ 0.47	14.85

MixiMotion uses 44.1% of the HY-Motion-1.0 full motion-generator parameters and one network evaluation rather than 50 solver evaluations. Under the same HY-Motion-family scope, it is $89.2\times$ faster for generation alone and $6.75\times$ faster end to end. The smaller end-to-end gain reflects shared text-encoding and motion-decoding costs. In the model-native profiles, MixiMotion’s one-step generator is $8.3\times$ faster than MotionHiFlow’s 14-evaluation hierarchical solver, and MotionLCM records the lowest latency and parameter count. Because their decoded endpoints differ, these cross-family latency values provide deployment context rather than a strict pipeline-level speedup claim. MotionLCM’s lower

SSAE score in Table 2 further shows that computational cost and semantic alignment should be considered together.

Human and qualitative evaluation.

Human evaluation. Table 4 reports the human ratings. HY-Motion-1.0 full has the highest mean score on all three criteria. MixiMotion is the strongest compact system, exceeding MotionHiFlow and MotionLCM on semantic alignment, naturalness, and overall motion quality. Its descriptive gap from the full baseline is 0.16 for semantic alignment and 0.17 for both naturalness and overall quality. The aggregate inter-rater agreement is Krippendorff’s $\alpha = 0.68$.

Table 4. Blinded human evaluation on a five-point Likert scale. Values are means over 300 prompts $\pm$ standard error. Higher is better. Bold denotes the best overall result and underlining denotes the best compact model.

Model	Semantic alignment	Naturalness	Overall motion quality
HY-Motion-1.0 full	4.48 ± 0.04	4.52 ± 0.03	4.50 ± 0.04
MixiMotion (ours)	<u>4.32 ± 0.05</u>	<u>4.35 ± 0.04</u>	<u>4.33 ± 0.05</u>
MotionHiFlow	4.18 ± 0.06	4.22 ± 0.05	4.20 ± 0.06
MotionLCM	3.95 ± 0.07	4.01 ± 0.06	3.98 ± 0.07

These descriptive results indicate that MixiMotion preserves much of the perceptual quality of the full HY-Motion-1.0 baseline while improving over the evaluated efficient baselines. They complement rather than replace the automated semantic and physical measurements.

Qualitative comparison. Following the qualitative presentation used by HY-Motion-1.0, we compare temporally ordered poses for three selected successful instructions in Fig. 3. Each prompt forms a separate panel, and the two HY-Motion-family outputs are rendered with identical geometry, camera, and lighting. The examples cover root translation, a sports action, and an upper-body social gesture. Across these cases, MixiMotion captures the principal action and its temporal progression, while the full model remains a strong reference for fine-grained pose execution. The cases were selected after visual inspection to illustrate successful behavior and must not be interpreted as a random or exhaustive sample. This visual comparison was not used to select checkpoints and does not replace the quantitative or blinded human evaluations.

4.3 Analysis

The main result is a quality–latency trade-off rather than an absolute quality win. MixiMotion retains 99.4% of the full HY-Motion-1.0 SSAE score while

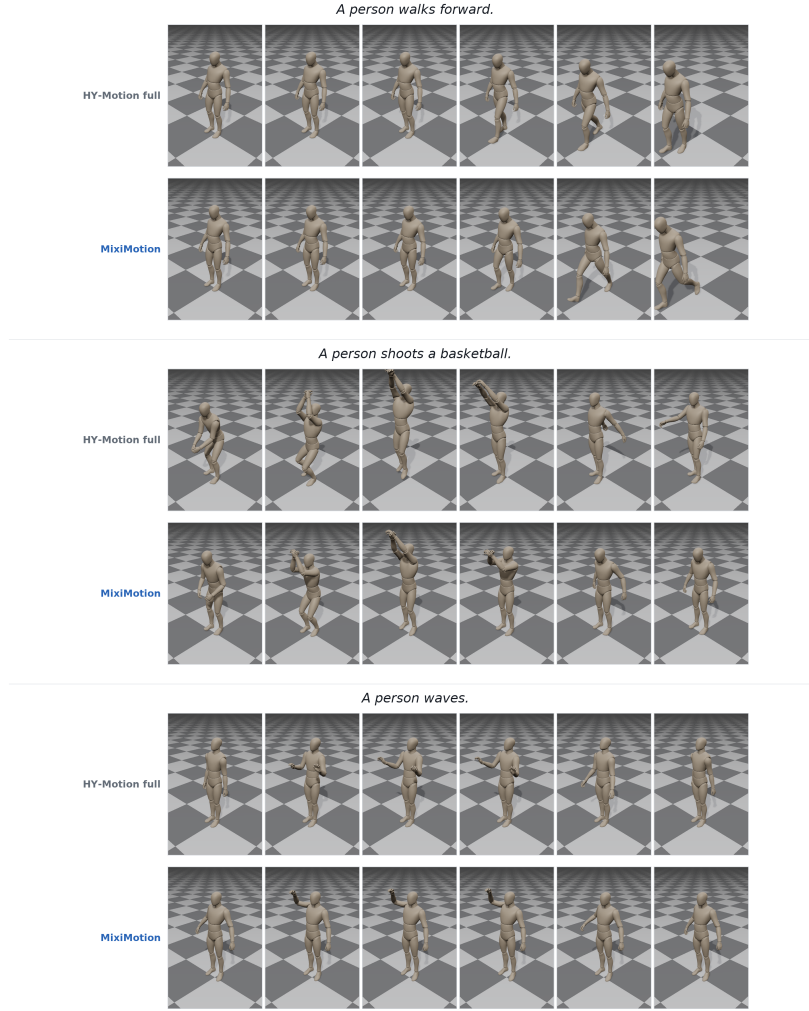


Fig. 3. Examples comparing HY-Motion-1.0 full (upper strip in each panel) and MixiMotion (lower strip). Each strip shows six matched frames in temporal order from the same 120-frame output. Early frames are shown for locomotion to keep the translating body in view.

using one generator evaluation and 44.1% of its motion-transformer parameters. The paired MixiMotion-minus-full interval $[-0.0121, 0.0011]$ includes zero, so the experiment does not resolve a difference between the two systems under this judge. It also does not establish equivalence: the interval still permits a nontrivial disadvantage, and the human means favor the full model by 0.16–0.17 points.

Efficiency depends strongly on measurement scope. Relative to the full model, the generator itself is $89.2\times$ faster, while the end-to-end gain is $6.75\times$ because text encoding and motion decoding remain. MixiMotion is also $8.3\times$ faster than MotionHiFlow at generation, yet its model-native end-to-end mean is higher (143.55 versus 83.80 ms) because the external baseline uses a smaller surrounding pipeline. Peak VRAM falls only from 23.23 to 20.73 GiB. MixiMotion therefore targets low sampling latency within the HY-Motion family; it is not the smallest or lowest-memory system in the table.

The category breakdown is more informative than the 0.0055 overall gap. MixiMotion ties the full model on locomotion and is within 0.0021 on game and sports, while its larger deficits occur on fitness, social, and daily actions. MotionHiFlow likewise exceeds MixiMotion on fitness and social. This pattern is consistent with the difficulty of compact one-step models in resolving fine-grained limb configuration, torso orientation, and interaction-like gestures. It should be treated as a hypothesis for targeted evaluation because SSAE categories mix many action types. The human means agree with the broad automatic ranking, but no paired significance test or participant-level confidence interval is reported; the human comparison is therefore limited to the observed sample means.

Teacher capacity. To test whether HY-Motion-1.0-Lite is the principal bottleneck, we repeat the complete Base–Quality–Semantic–DPO schedule after regenerating all cached targets with HY-Motion-1.0 full. The two runs use the same 48,065 prompt rows after exact-text filtering, four teacher candidates per prompt, the same 460.0M-parameter student architecture, curriculum configuration, rendering protocol, and native-video judge. Only the model used to create the offline teacher cache changes.

Table 5. Teacher-capacity diagnostic for the final one-step student. Both teacher banks have zero exact normalized-text overlap with SSAE. Higher SSAE is better.

Offline cache teacher	Cache rows	Student NFE	Overall SSAE
HY-Motion-1.0-Lite	48,065	1	0.8802
HY-Motion-1.0 full	48,065	1	0.8796

The larger teacher does not improve the aggregate score under this recipe: its student is lower by 0.0006. It is numerically stronger on fitness (0.7658 vs. 0.7487) and locomotion (0.9786 vs. 0.9757), but weaker on daily, game, social, and sports prompts. This single retraining does not establish equivalence between teachers and is not a direct evaluation of HY-Motion-1.0-Lite itself. It instead shows that increasing offline teacher capacity alone does not remove the present one-step bottleneck.

Because there is only one run per teacher, the 0.0006 difference is too small to identify a teacher-capacity effect. Candidate coverage with $K = 4$, student

capacity, one-step optimization, lexical group definitions, and the distance-based preference surrogate remain plausible bottlenecks.

Ablation study. Before clean retraining, we used one cumulative checkpoint chain to inspect the effect of each curriculum stage. All rows use the same architecture, teacher cache, rendering pipeline, and native-video judge, and each row inherits every preceding stage. We additionally report foot-skating and contact metrics for each checkpoint. Because these runs were inspected during development and predate exact-text filtering, Table 6 is a training diagnostic rather than a held-out component ablation. Its values are not mixed with the clean-retrained checkpoint in the primary comparison.

Table 6. Cumulative checkpoint diagnostic from the preceding development run. Higher SSAE and contact scores are better; lower foot skating is better. Physical metrics are dataset means over the 1,969 development motions. Best values are in bold.

Variant	Native-video SSAE	Foot skating	Contact score
Base distillation	0.8822	0.0401	0.7901
+ Motion-quality	0.8802	0.0495	0.7621
+ Semantic balance	0.8824	0.0589	0.7376
+ Preference/DPO (development)	0.8813	0.0558	0.7513

Base distillation and semantic balancing obtain nearly identical SSAE scores. The motion-quality and final development preference checkpoints are lower than the base by 0.0020 and 0.0009, respectively. The aggregate physical metrics are also non-monotonic: later stages have more foot skating and less contact than the base, while preference optimization partly reverses the changes introduced by semantic balancing. Thus, this ablation does not support a claim that every curriculum stage improves these aggregate metrics. For the development DPO checkpoint, the foot-skating ratio is 0.0558 ± 0.1433 and the contact ratio is 0.7513 ± 0.3347 across 1,969 motions. The same renderer-side implementation and thresholds are used for every checkpoint.

A plausible explanation is that group-specific gait and stationary objectives redistribute errors across prompt types in ways that dataset-level averages hide. Establishing a benefit requires a clean stage-wise ablation with per-group physical metrics and repeated seeds. The preference objective follows the reference-relative logistic construction of Rafailov et al., but substitutes negative best-of- M distances for model log probabilities. It therefore favors proximity to a physically selected teacher winner relative to a loser; it does not directly optimize SSAE, human preference, or calibrated sequence likelihood. The non-monotonic development result is compatible with that objective and motivates direct preference validation.

5 Limitations and Future Work

Exact normalized-text filtering removes direct SSAE prompt copies, while semantic paraphrases and near-duplicates may remain. All automatic semantic scores use one deterministic VLM judge, and the external compact baselines retain different renderers and frame rates. The blinded human study covers 25 participants and 300 prompts per system, reports standard errors and aggregate agreement, and lacks participant-level inferential comparisons. The qualitative cases are explicitly selected successful examples. Finally, all motions have a fixed 120-frame length, and the evaluation does not directly measure sample diversity or distributional fidelity. These constraints support a controlled HY-Motion-family efficiency claim and a descriptive compact-model comparison, while leaving broader benchmark generalization unresolved.

6 Conclusion

We presented MixiMotion, our one-step HY-Motion-1.0 model trained with a progressive curriculum distillation recipe for text-to-motion generation. MixiMotion uses one-step inference and 460.0M rather than 1.04B motion-generator parameters. After retraining on a teacher bank with zero exact normalized-text SSAE overlap, MixiMotion obtains 0.8802 SSAE versus 0.8856 for HY-Motion-1.0 full in the all-frame native-video evaluation, retaining 99.4% of its score and obtaining the highest numerical score among the evaluated compact models. Public-baseline comparisons remain contextual because their native renderers and frame rates differ. The method provides a practical route toward interactive text-to-motion systems. The blinded human study likewise places MixiMotion above the evaluated compact baselines and slightly below HY-Motion-1.0 full. A matched full-teacher retraining does not improve the aggregate one-step score, indicating that teacher scale alone is not sufficient. Evaluation on an independently sourced benchmark and a common cross-family renderer remains important for establishing broader generalization.

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